

How Many Words Do I Know in My Native Mandarin?

LING 5832 — Assignment 1, Part 1 • Yiyuan Jia • August 31, 2026

1 The short answer

My native language is Mandarin Chinese. My answer: **I know approximately 60,800 of the 70,414 headword entries of 《现代汉语词典》 (*Contemporary Chinese Dictionary*, 7th edition; hereafter XHC7), with a 95% confidence interval of 58,400–63,200 — about 86% of the standard reference dictionary of Modern Standard Chinese.** I obtained this by stratified random sampling of the dictionary’s full text and a two-stage self-test in which every claim of knowledge was checked against the revealed definition. The single number, however, hides three definitional choices — what a *word* is (unusually fraught in Mandarin), what *knowing* one means, and what population “how many” quantifies over. Depending on those choices my defensible answers range from about 6,700 (single-character morpheme entries, strictly known) to 65,600 (all entries, trusting self-report); this report makes the choices explicit.

2 What counts as a “word”? Mandarin morphology and the unit problem

Mandarin makes “word” hard to pin down. The orthography marks no word boundaries — this is why Chinese NLP needs a word-segmentation step at all — and the culturally salient unit is the 字 *zì* (character), which corresponds roughly to a morpheme-syllable rather than to a word. The linguistic word, 词 *cí*, typically spans one to four characters, and native-speaker intuitions about where 词 boundaries fall are famously unstable (Packard 2000).

Morphologically, Mandarin is strongly isolating: no agreement, case, or tense; only a few aspect clitics (了, 着, 过) and the restricted plural 们. The lemma-versus-wordform decision that can shift an English count by a factor of two, and a Finnish or Turkish count by far more, therefore barely arises. What replaces it is word *formation*: the modern lexicon is built overwhelmingly by compounding (mostly disyllabic, e.g. 词典 ‘word-dictionary’ = dictionary), with some derivation (老-, 阿-, -子, -化, -性) and reduplication (看看, 高高兴兴). Where an English speaker’s vocabulary-size question is mostly “how many lemmas have you memorized,” a Mandarin speaker’s is partly “how productive is your compounding” — and that second question has no finite answer.

I therefore operationalized *word* as **headword entry in XHC7**: the dictionary’s own individuation of the lexicon (written form + pronunciation + sense set), 70,414 entries in the full-text transcription. This is countable and reproducible, but it inherits lexicographic compromises. First, 11,424 entries (16%) are single-character 字头, many of them bound morphemes rather than free words; of the ones I did not know, most were archaic literary characters such as 垌 ‘open country’ and 屐 ‘shoe.’ Second, 6,497 four-character entries are mostly 成语 — lexicalized idioms that are structurally phrases. Third, separable verb-object items (离合词) such as 睡觉 ‘sleep,’ which split freely (睡了一个好觉), blur the word–phrase line. Fourth, 2,334 written forms map to more than one entry (啊 has five, one per tone or function), so entry-counting splits homographs. Finally, transparent compounds are simply absent: XHC7 lists 鸭蛋 ‘duck egg’ but not 鸡蛋 ‘hen’s egg,’ 牙刷 ‘toothbrush’ but not 电动牙刷 ‘electric toothbrush.’ I know the unlisted member of each pair exactly as well as the listed one, but only the listed one is countable — the dictionary records what is lexicalized, not what I can understand or produce.

3 What counts as “knowing” a word?

“Know” spans a gradient from “the form looks familiar” to “I can deploy it accurately.” I operationalized it as a two-stage judgment made for every sampled item, with the bare written form as the stimulus — no pinyin, no gloss — so that retrieving pronunciation and meaning was part of the test. **Stage 1**, self-rating on a four-point scale: 不认识 (don’t know it) / 眼熟 (form familiar, cannot retrieve a meaning) / 认识 (can read it and give the approximate meaning) / 掌握 (could use it correctly in my own speech or writing). **Stage 2**, verification: for every claim of 认识 or 掌握, the dictionary definition was revealed and I judged my prior understanding 对 (correct) / 部分对 (partially correct) / 错了 (wrong).

Verification is the guard against the well-known inflation of self-reported vocabulary, and it caught me 25 times in 336 claims (7.4%): 13 outright wrong — including, embarrassingly for a computational-linguistics student, 黏着语

‘agglutinative language’ — and 12 partially right, concentrated in semi-transparent 成语 such as 集腋成裘, whose parts suggest a meaning that is only approximately the lexicalized one. Verification also disambiguates homographic forms: claiming to know a character while thinking of a different pronunciation’s entry surfaces as “wrong” at this stage. From the judgments I define four operationalizations of “know”: **receptive-strict** (rated 认识 or better, verified 对) — the headline; **receptive-lenient** (verified 对 or 部分对); **productive** (rated 掌握 and verified 对); and **self-report** (认识 or better, ignoring verification), kept to measure the inflation. Response times, logged incidentally, corroborate the scale: median latency was 674 ms for 掌握 judgments versus 2.3–3.6 s for all other categories — the signature of automatic lexical access.

4 “How many”: estimation, not enumeration

No one can enumerate their lexicon, so I interpret “how many” as a point estimate, with quantified sampling uncertainty, of a well-defined population quantity: the number of XHC7 entries I know. I parsed the full text into 70,414 entries (excluding front matter and the appendix of Western-alphabet items such as NBA), stratified by headword length, and drew a seeded random sample of 360 items: 60/180/45/55/20 from the 1/2/3/4/5+-character strata (population sizes 11,424 / 45,086 / 6,690 / 6,497 / 717). The long-item strata are deliberately oversampled so that the 成语 and 熟语 proportions are estimable; the stratified weighting corrects for this. The estimator is the standard stratified total $\sum N_h \hat{p}_h$ with finite-population-corrected standard errors; because several strata had $\hat{p}$ at or near 1, where the normal approximation is anti-conservative, I also report a more cautious interval built from per-stratum Wilson bounds. Presentation order was randomized across strata; the test took about 25 minutes on August 31, 2026.

5 Results

Table 1. Per-stratum results under the receptive-strict definition.

Stratum	N _h	n _h	known	$\hat{p}_h$	est. known
1 character (字头)	11,424	60	35	.583	6,664
2 characters	45,086	180	166	.922	41,579
3 characters	6,690	45	41	.911	6,095
4 characters (mostly 成语)	6,497	55	49	.891	5,788
5+ characters (熟语/谚语)	717	20	20	1.000	717
Total	70,414	360	311	—	60,844

Table 2. Estimated entries known under four operationalizations of “know.”

Definition	Estimate	95% CI (normal)	Wilson-summed
Self-report only	65,604	[63,924, 67,284]	[61,587, 67,426]
Receptive, lenient	63,140	[61,032, 65,248]	[58,316, 65,991]
Receptive, strict	60,844	[58,441, 63,247]	[55,599, 64,299]
Productive	59,581	[57,085, 62,078]	[54,270, 63,215]

Knowledge is near ceiling for multi-character words — 92% of two-character entries, 91% of three-character entries, 89% of sampled 成语, and 20 of 20 longer sayings — and far lower (58%) for single-character 字头, exactly the stratum where the dictionary stocks archaic, literary, and dialectal bound morphemes. Restricting to multi-character entries (“words” in the everyday sense) gives 54,180 of 58,990, or 92%. Verification matters: it removed about 4,800 entries (7%) from the self-report estimate, and the strict-lenient gap (how 部分对 is scored) moves the answer by another ±2,300.

6 Where the method and the definitions come apart

1. **The dictionary is not my lexicon, in both directions.** XHC7 (2016) omits my technical vocabulary, recent neologisms, proper names, and all transparent compounds I understand and produce freely — an undercount of “words I know”; yet it counts bound-morpheme 字头 and splits homographic forms into several entries — an overcount of “words.” The estimate is best read as my *coverage of a curated reference lexicon*, a proxy for, not a measure of, the size of my mental lexicon.
2. **“Productive” was never tested by production.** 掌握 is still a self-assessment; an actual production task (write a sentence, have it judged) would likely lower the 59,581.
3. **Verification is sense-coarse.** 对 requires only one matching sense of a polysemous entry, so entry-level “knowing” overstates sense-level knowledge; a sense-level study would use a much larger denominator (XHC7 distinguishes numbered senses per entry).
4. **Transcription gaps.** About 4,200 entries (6%) of the community transcription — mostly 7th-edition additions — lack transcribed definitions; 25 of my 336 claims hit such blanks, so for them stage 2 silently degenerated to self-report. Nearly all were common items (中国象棋, 老伴儿), so the bias is small but real.
5. **One subject, one session.** The confidence intervals quantify sampling error only; they say nothing about the stability of my own judgments (no test–retest, no second rater).
6. **Cross-language comparison is treacherous.** Estimates like 17,000 word families for educated English natives (Goulden, Nation & Read 1990) are not commensurable with 60,800 XHC7 entries: the units differ (families collapse derivational kin; entries split homographs and include idioms) and so do the universes (XHC7 curates ~70k entries; *Webster’s Third* has ~267k).

7 Conclusion

How many words do I know in Mandarin? About **61,000 dictionary entries** — six of every seven in the standard reference dictionary — under a receptive criterion with definition-checking. But the more instructive answer is the decomposition that Mandarin’s morphology forces: I strictly know about 6,700 of the 11,424 single-character morpheme entries, essentially all of the high- and mid-frequency compound vocabulary, roughly nine in ten 成语 — and, beyond anything countable, an open set of transparent compounds that no dictionary will ever list, because the morphology builds them on demand. “How many words do you know” has a numeric answer only after you decide what a word is; in Mandarin, that decision is most of the question.

Reproducibility: the parsing, seeded sampling (seed 5832), self-test app, analysis scripts, and my 360 raw judgments accompany this report in the assignment repository; the dictionary text is the community transcription at github.com/CNMan/XDHYCD7th.

References

- Goulden, R., Nation, P., & Read, J. (1990). How large can a receptive vocabulary be? *Applied Linguistics*, 11(4), 341–363.
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